Assignment 2

Instructions

Attempt the following questions. Show all work and reasoning to receive full credit.

Questions

1. Given the following matrix, find its row echelon form:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ 3 & 6 & -3 \end{bmatrix}.$$

   Is the matrix in row echelon form already? Justify your answer.

2. For the matrix

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix},$$

   find the reduced row echelon form. Explain each row operation used.

3. Determine if the following set of vectors is linearly independent:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 3 \\ 6 \\ 3 \end{bmatrix}.$$

   If they are dependent, find a linear relationship among them.

4. Determine if the vector $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ can be written as a linear combination of the vectors

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

   If it can be, express $\mathbf{b}$ as a linear combination of $\mathbf{u}$ and $\mathbf{v}$.

5. Consider the set $S = \{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \}$. Does S span $\mathbb{R}^3$? Justify your answer.

6. Determine if the set

$$T = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \mid x + 2y - z = 0 \right\}$$

is a subspace of $\mathbb{R}^3$. If it is, show that it satisfies the properties of a vector space.

7. Let V be the vector space of all polynomials of degree at most 2. Determine if the subset

$$W = \{a + bx \mid a, b \in \mathbb{R}\}$$

is a subspace of V. Justify your answer.

8. Find a basis for the null space of the matrix

$$C = \begin{bmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \\ -1 & 1 & -2 \end{bmatrix}.$$

9. For the vector space $\mathbb{R}^4$, find a basis for the span of the following vectors:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \\ 8 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 3 \\ 6 \\ 9 \\ 12 \end{bmatrix}.$$

10. Let $U = \left\{ \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \in \mathbb{R}^4 \mid x - y + z = 0 \text{ and } y + w = 0 \right\}$. Is U a subspace of $\mathbb{R}^4$? If so, find a basis for U.